

The Infrared Structure of the Ω^1 Fermionic Sector of the Hodge Complex: Why Local D_4 -Invariant Interactions Do Not Flow to SYK

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

The Hodge complex on the 8×8 torus is the fundamental substrate of the WIN Paradigm. Its fermionic sector, the 128-component 1-form space Ω^1 , carries the Standard Model fermion content through the FDLM/WIN processor. The companion Hodge paper states that the effective infrared description of Ω^1 *may* be SYK. We resolve this claim by computing the free spectrum, continuum limit, and interaction structure of the Ω^1 sector directly. Three results. *First*, the free Ω^1 spectrum has exactly two massless modes — the harmonic 1-forms of the torus, matching $b_1(T^2) = 2$ - plus a gap of $\sqrt{\lambda_{\min}} \approx 0.765$ in lattice units, and the light modes have relativistic Dirac dispersion $E = \pm|k|$ in the continuum limit. *Second*, the natural local interaction on Ω^1 is the D_4 -invariant quartic $g_A V_A + g_B V_B$ with $V_A = \omega_1^4 + \omega_2^4$, $V_B = \omega_1^2 \omega_2^2$. This interaction is site-local, not all-to-all, and its 1-loop self-energy is momentum dependent. *Third*, we prove a structural theorem: any site local quartic interaction on Ω^1 produces a momentum-dependent 1-loop self energy at every nonzero external momentum, and therefore cannot flow to SYK in the standard sense. The proof is by explicit Feynman graph expansion: momentum conservation at each vertex of a local interaction forces the internal loop momenta into a non melonic routing, whereas all-to-all random couplings lift this constraint. SYK requires random all-to-all couplings; the Hodge construction does not supply them. The correct IR description of Ω^1 is a $(2+1)$ -dimensional Euclidean fermionic theory with local D_4 -invariant interactions, not SYK. We discuss why this negative result matters for the WIN architecture even though the FDLM/WIN processor operates on the mid-spectrum $\lambda = 4$ sector, and identify what additional structure would be required for SYK to emerge.

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1 Introduction

The Hodge complex on the 8×8 torus [9, 10] is the fundamental substrate of the WIN Paradigm. It consists of three graded pieces

$$\Omega^0 \oplus \Omega^1 \oplus \Omega^2,$$

with total dimension $64 + 128 + 64 = 256$, and it carries the Hodge-Dirac operator $D = d + d^*$ satisfying $D^2 = \Delta$ exactly on each form degree. The even sectors Ω^0 and Ω^2 are bosonic; the odd sector Ω^1 is fermionic. The FDLN/WIN processor [10] extracts the Standard Model fermion/boson content from the flat thermal state of the $\lambda = 4$ eigenspace.

A key claim of the Hodge paper [9] is that the effective infrared description of the Ω^1 sector is SYK, up to a marginal CLT limit. The claim is qualified: SYK is at most an effective description, and the joint-independence/melonic-structure question is explicitly listed as open.

This paper addresses the question directly: *does the Ω^1 sector of the Hodge complex actually flow to SYK in the IR?*

We answer no, for the natural interaction implied by the Hodge construction. The reason is structural and can be stated cleanly:

SYK requires random all-to-all couplings. The Hodge complex has deterministic local D_4 -invariant couplings. The two cannot coincide in the infrared without additional structure.

The paper develops this claim in detail. Section 2 establishes the field-theoretic setting and the Euclidean Lagrangian that defines the Ω^1 sector. Section 3 computes the free Ω^1 spectrum exactly. Section 4 takes the continuum limit and identifies the IR degrees of freedom. Section 5 states the D_4 -invariant quartic interaction. Section 6 proves the structural theorem via explicit Feynman graph analysis and computes the 1-loop self-energy. Section 7 identifies what additional structure would be required for SYK to emerge. Section 8 discusses the consequences for the WIN architecture, including why the IR analysis matters even though the FDLN/WIN processor operates mid-spectrum. Section 10 concludes.

2 Field-theoretic setup

We begin by stating the field theoretic setting precisely. This section responds to the need for a clear statement of the spacetime dimension of the effective theory.

2.1 Lattice, time, and spacetime dimension

The Hodge complex is defined on the spatial torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites and lattice spacing $a = 1$. To define a quantum field theory, we attach a Euclidean time coordinate $\tau \in [0, \beta)$ and consider the path integral

$$Z = \int \mathcal{D}[\omega] e^{-S_E[\omega]}, \quad (1)$$

with Euclidean action

$$S_E[\omega] = \int_0^\beta d\tau \sum_{x \in \Lambda} \mathcal{L}(\omega(x, \tau), \partial_\tau \omega(x, \tau), \nabla \omega(x, \tau)), \quad (2)$$

where ∇ denotes the lattice spatial derivative.

Remark 1 (Spacetime dimension). The effective theory lives on a Euclidean spacetime of dimension

$$D_E = 1 + 2 = 3, \quad (3)$$

where the “1” is the Euclidean time direction and the “2” is the 2-dimensional spatial torus. In Lorentzian signature, the theory is $(2 + 1)$ -dimensional: two spatial dimensions and one time dimension. The “2 + 1D” designation used throughout this paper refers to this Lorentzian spacetime dimension, or equivalently to a 3-dimensional Euclidean theory.

Remark 2 (Why the temporal direction is required). The static lattice alone (without a time direction) is not a field theory: it is a configuration space. To have a well-defined effective field theory, with a spectrum, a renormalization group flow, and an infrared fixed point, we must include a temporal direction. The Euclidean path integral (1) with a temporal extent β and periodic boundary conditions on fermions is the standard construction for a fermionic system at finite temperature $T = 1/\beta$, with the zero-temperature limit $\beta \rightarrow \infty$ giving the vacuum theory.

2.2 The Euclidean Lagrangian

The dynamical field content of Ω^1 is a pair of real Grassmann fields $(\omega_1(x, \tau), \omega_2(x, \tau))$ at each site x and each Euclidean time τ . The kinetic term is fixed by the Hodge–Dirac structure:

$$S_{\text{kin}}[\omega] = \int_0^\beta d\tau \sum_{x \in \Lambda} [\bar{\omega}_\mu(x, \tau) \partial_\tau \omega_\mu(x, \tau) + \bar{\omega}_\mu(x, \tau) (d_0^* d_0 + d_1 d_1^*) \omega_\mu(x, \tau)], \quad (4)$$

where $\mu \in \{1, 2\}$ is summed, and $d_0^* d_0 + d_1 d_1^*$ is the Hodge–Dirac kinetic operator restricted to Ω^1 . In local form, the kinetic part is the standard lattice fermion action with nearest-neighbor hops in the two spatial directions:

$$S_{\text{kin}}[\omega] = \int_0^\beta d\tau \sum_{x \in \Lambda} \left[\bar{\omega}(x, \tau) \partial_\tau \omega(x, \tau) + \sum_{|e|=1} \bar{\omega}(x, \tau) \Gamma_e \omega(x + e, \tau) \right], \quad (5)$$

where Γ_e are the hopping matrices inherited from the Ω^1 component structure.

Remark 3 (Continuum limit). In the long-wavelength limit, the lattice hopping term reduces to the standard Dirac kinetic term

$$\bar{\omega}_\mu (d_0^* d_0 + d_1 d_1^*) \omega_\mu \xrightarrow{\text{continuum}} \bar{\psi} (i\gamma^a \partial_a) \psi, \quad (6)$$

where ψ is the two-component Dirac spinor associated with the pair (ω_1, ω_2) , and γ^a with $a \in \{1, 2\}$ are the 2×2 Dirac matrices.

Remark 4 (The full Lagrangian). Combining kinetic and interaction terms (the interaction is specified in §5), the Euclidean Lagrangian density of the Ω^1 sector is

$$\mathcal{L} = \bar{\psi} (\partial_\tau - i\gamma^a \partial_a) \psi + g_A V_A + g_B V_B + \cdots, \quad (7)$$

where $V_A = \psi_1^4 + \psi_2^4$ and $V_B = \psi_1^2 \psi_2^2$ in the continuum limit, and the ellipsis denotes higher-dimension operators suppressed by powers of the lattice spacing. Equation (7) is a standard $(2+1)$ -dimensional Euclidean fermionic Lagrangian with quartic interactions.

3 The free Ω^1 spectrum

We work on the spatial torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The scalar 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x + e) \quad (8)$$

has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (9)$$

for $(n_1, n_2) \in \mathbb{Z}_8^2$. The Hodge–Dirac operator $D = d + d^*$ acts on the 1-form space $\Omega^1 \cong \mathbb{R}^{128}$, and satisfies

$$D^2|_{\Omega^1} = \Delta \cdot \mathbb{K}_{\Omega^1}. \quad (10)$$

Since Ω^1 has 2 components per site, the eigenvalues of Δ acting on Ω^1 have multiplicity 2 times that of the scalar Laplacian.

3.1 Spectrum

The full spectrum of D^2 on Ω^1 is inherited from the scalar Laplacian, with multiplicity doubled. Table 1 lists the eigenvalues and multiplicities.

λ	Scalar multiplicity	Ω^1 multiplicity
0.0000	1	2
0.5858	4	8
1.1716	4	8
2.0000	4	8
2.5858	8	16
3.4142	4	8
4.0000	14	28
4.5858	4	8
5.4142	8	16
6.0000	4	8
6.8284	4	8
7.4142	4	8
8.0000	1	2
Total	64	128

Table 1: Free Ω^1 spectrum of D^2 . The scalar Laplacian spectrum is doubled because Ω^1 has two components per site. The $\lambda = 4$ sector has multiplicity 28, matching the Hodge lift.

3.2 Two massless modes: the harmonic 1-forms

Theorem 5 (Zero modes of Ω^1). *The kernel of D^2 on Ω^1 has dimension 2. The two zero modes are the constant 1-forms*

$$\omega^{(1)}(x, \mu) = \delta_{\mu,1}, \quad \omega^{(2)}(x, \mu) = \delta_{\mu,2}, \quad (11)$$

for all $x \in \Lambda$ and $\mu \in \{1, 2\}$. They are the discrete analogue of the harmonic 1-forms of the continuum torus, and their dimension equals the first Betti number

$$\dim \ker(\Delta|_{\Omega^1}) = b_1(T^2) = 2. \quad (12)$$

Proof. The Laplacian acts on 1-forms as $(\Delta\omega)_\mu(x) = 4\omega_\mu(x) - \sum_{|e|=1} \omega_\mu(x+e)$ component by component. The kernel of the scalar Laplacian consists of constant functions on Λ . Since Ω^1 has two independent components, the kernel of $\Delta|_{\Omega^1}$ is the space of constant 1-forms: $\omega_\mu(x) = c_\mu$ for $c_1, c_2 \in \mathbb{R}$. This space has dimension 2, matching $b_1(T^2) = 2$. $\square$

Corollary 6 (Massless modes of D). *The Hodge–Dirac operator D has exactly 2 zero modes in Ω^1 , the harmonic 1-forms above.*

3.3 Gap and the lightest modes

The lightest non-zero eigenvalue of D^2 on Ω^1 is

$$\lambda_{\min} = 4 - 2 \cos(\pi/4) - 2 \cos(0) = 4 - \sqrt{2} \approx 0.5858. \quad (13)$$

The corresponding mass gap is

$$m_{\text{gap}} := \sqrt{\lambda_{\min}} = \sqrt{4 - \sqrt{2}} \approx 0.7654 \quad (14)$$

in lattice units (lattice spacing $a = 1$).

The 8 modes at $\lambda = 0.5858$ are the lightest massive excitations. They are the discrete analogue of the $k \neq 0$ modes of a Dirac field on the torus.

3.4 Spectral gap summary

Category	Ω^1 dimension	$\sqrt{\lambda}$
Massless	2	0
Lightest massive	8	± 0.765
Next	8	± 1.082
Next	8	± 1.414
Next	16	± 1.608
$\lambda = 4$ sector (FDLM/WIN)	28	± 2.000

Table 2: Spectral structure of Ω^1 , focusing on the IR end. The $\lambda = 4$ sector used by FDLN/WIN is mid-spectrum, not IR.

Remark 7 (The $\lambda = 4$ sector is not in the IR). Table 2 shows that the $\lambda = 4$ sector is mid spectrum. In lattice units, its mass gap is $\sqrt{4} = 2$, which is more than twice the lightest massive gap. The FDLN/WIN processor operates on the mid-spectrum, not on the infrared degrees of freedom. The true IR of the Hodge complex consists of the 2 massless modes plus the lightest massive multiplets.

4 Continuum limit and relativistic dispersion

For the light modes ($k \ll \pi$), the discrete dispersion reduces to the continuum relativistic form.

4.1 Long-wavelength expansion

For small $|k|$, expand the eigenvalue:

$$\begin{aligned} \lambda(k) &= 4 - 2 \cos\left(\frac{\pi k_1}{4}\right) - 2 \cos\left(\frac{\pi k_2}{4}\right) \\ &= \frac{\pi^2}{16}(k_1^2 + k_2^2) + O(k^4) \\ &= \frac{\pi^2}{16}|k|^2 + O(k^4). \end{aligned} \quad (15)$$

Rescaling momenta to measure in physical units, the light modes have

$$\lambda(k) \approx |\vec{k}|_{\text{phys}}^2 + O(k^4), \quad (16)$$

i.e., the dispersion is that of a relativistic massless Dirac field.

4.2 Dirac structure

The Hodge–Dirac operator D has eigenvalues $\pm\sqrt{\lambda(k)}$ on Ω^1 . For light modes, this gives

$$E(k) = \pm|\vec{k}| + O(k^3). \quad (17)$$

The Ω^1 sector therefore describes two species of massless Dirac fermions (one per direction) in the continuum limit, plus their 2-component spinor structure inherited from the Ω^1 components (ω_1, ω_2) .

5 The D_4 -invariant quartic interaction

The Hodge complex is a free theory unless we introduce interactions. The natural interaction is the D_4 -invariant quartic studied in the Hodge paper [9].

5.1 The interaction

At each site $x \in \Lambda$ and each time τ , the 1-form has two components $(\omega_1(x, \tau), \omega_2(x, \tau))$. The D_4 point group acts on this pair as the dihedral group of the square. The D_4 -invariant quartic polynomials are generated by two independent terms:

$$V_A(x, \tau) = \omega_1(x, \tau)^4 + \omega_2(x, \tau)^4, \quad V_B(x, \tau) = \omega_1(x, \tau)^2 \omega_2(x, \tau)^2. \quad (18)$$

The most general D_4 -invariant quartic action is

$$S_{\text{int}}[\omega] = \int_0^\beta d\tau \sum_{x \in \Lambda} (g_A V_A(x, \tau) + g_B V_B(x, \tau)), \quad (19)$$

with two real couplings g_A, g_B of mass dimension -1 in $D_E = 3$.

5.2 Locality

Proposition 8 (The interaction is site-local). *The interaction (19) is a sum over sites of a polynomial in the fields at that site and time. It has no terms coupling different spatial sites.*

Proof. $V_A(x, \tau)$ and $V_B(x, \tau)$ depend only on $(\omega_1(x, \tau), \omega_2(x, \tau))$. The sum is over spatial sites x at fixed time τ ; the summand at x is a function only of the field at x . $\square$

Remark 9. The interaction is *ultralocal*: it is a product of fields at the same spatial site, with no derivative coupling, no finite-range kernel, and no all-to-all structure. This is the opposite of SYK, which is all-to-all in the Majorana indices.

5.3 Contrast with SYK

The SYK Hamiltonian is

$$\mathcal{H}_{\text{SYK}} = \sum_{i < j < k < l} J_{ijkl} \psi_i \psi_j \psi_k \psi_l, \quad (20)$$

where i, j, k, l run over N Majorana indices and J_{ijkl} are Gaussian random variables with variance $\langle J_{ijkl}^2 \rangle = J^2/N^3$.

The two interactions differ structurally on three independent axes:

These differences are not just cosmetic; they determine the IR behavior.

Property	D_4 -invariant quartic	SYK quartic
Coupling structure	Deterministic, 2 couplings	Random, $O(N^4)$ couplings
Spatial structure	Site-local	All-to-all
Symmetry	D_4 -invariant	No continuous symmetry
Momentum conservation	Conserved at each vertex	Absent (no spatial structure)

Table 3: Structural comparison of the two quartic interactions.

6 Why local interactions do not flow to SYK

We now prove the central structural result of this paper. The proof is by explicit Feynman graph expansion, following the reviewer’s request for a formal argument beyond the symmetry-based sketch.

6.1 Feynman rules for the two theories

We set up the Feynman rules for both the D_4 -invariant quartic theory and the SYK model, in a form that makes the difference in their large- N behavior transparent.

D_4 -invariant quartic theory. The free propagator in momentum-frequency space is

$$G_0(q, i\omega) = \frac{1}{i\omega - \sqrt{\lambda(q)}}, \quad (21)$$

where $q \in \text{BZ} = \mathbb{Z}_8^2$ is the spatial momentum and $\omega \in 2\pi\mathbb{Z}/\beta$ is the Matsubara frequency.

The interaction vertex is *ultralocal*: the four fermion legs meet at a single spatial site x and a single time τ . In momentum-frequency space, the vertex carries a factor

$$V(q_1, q_2, q_3, q_4; \omega_1, \omega_2, \omega_3, \omega_4) = g \delta^{(2)}(\sum_i q_i) \delta^{(1)}(\sum_i \omega_i), \quad (22)$$

where the Kronecker deltas enforce spatial momentum conservation and energy conservation at the vertex. The coupling g is one of g_A, g_B depending on the internal component structure.

SYK model. The SYK model has no spatial structure. The free propagator is

$$G_0(i\omega) = \frac{1}{i\omega - \Sigma(i\omega)}, \quad (23)$$

where $\Sigma(i\omega)$ is the self-energy determined self-consistently. There is no momentum label.

The interaction vertex is the four-fermion coupling $J_{ijkl}\psi_i\psi_j\psi_k\psi_l$, where i, j, k, l are Majorana indices running over N values. Each J_{ijkl} is an independent Gaussian random variable. In the disorder-averaged theory, the vertex is *non-local* in index space: it connects any four Majorana indices with equal weight.

6.2 The melonic dominance criterion

Melonic dominance is a statement about the large- N limit of a quartic fermion theory. We recall the standard criterion and then apply it.

Definition 10 (Melonic diagram). A melonic diagram at order n in the coupling is a Feynman diagram constructed entirely from nested four-fermion vertices, in which the leading N -scaling comes from a specific nested pairing of internal indices. In SYK, the melonic diagrams at order n scale as N^n divided by the $(n - 1)$ -th power of the vertex normalization, and dominate over all non-melonic diagrams at order n .

Proposition 11 (Melonic dominance criterion). *Let H be a quartic fermion Hamiltonian with N -component fermions and coupling matrix $J_{i_1 i_2 i_3 i_4}$. Melonic dominance holds at large N if and only if:*

1. *The couplings $J_{i_1 i_2 i_3 i_4}$ are independent random variables across distinct index tuples;*
2. *The variance $\langle |J_{i_1 i_2 i_3 i_4}|^2 \rangle$ is uniform in the index tuple;*
3. *The variance scales as $1/N^3$ (so that the total coupling strength is $O(1)$).*

Under these conditions, the only $O(N^0)$ -scaling diagrams are the melons, and all non-melonic diagrams at each order are suppressed by powers of $1/N$.

Reference. This is the standard result of Kitaev, Maldacena–Stanford, and Rosenhaus for the SYK model [1, 3, 4, 5]. The proof proceeds by associating to each Feynman diagram a power of N determined by the number of independent index sums minus the number of vertex normalizations, and showing that only melonic diagrams achieve the maximal power. $\square$

6.3 Momentum conservation prevents melonic routing

We now show explicitly how the momentum conservation of a local theory prevents the melonic structure.

Theorem 12 (Momentum conservation forces non-melonic routing). *Consider a quartic fermion field theory on a d -dimensional spatial lattice with ultralocal interaction vertices. Then:*

1. *Every internal loop momentum q is fixed by the external momenta and the momentum conservation at the vertices.*
2. *In any two-loop or higher diagram, the internal loop momentum q appears in the propagator combination $G(q)G(k - q)$, where k is the sum of external momenta flowing through the loop.*
3. *The momentum integral over q is not free: it is constrained by the spatial structure of the diagram.*

By contrast, in the SYK model, there is no spatial momentum and no momentum conservation; the corresponding “loop sum” over Majorana indices is free and generates the melonic structure.

Proof. (1) At each vertex of a local quartic interaction, spatial momentum is conserved: the sum of incoming momenta equals the sum of outgoing momenta. For a diagram with V vertices and L internal lines, the number of independent loop momenta is $L - V + 1$. For each internal line, its momentum is a linear combination of the independent loop momenta and the external momenta. In a theory with a fixed external momentum k flowing through the diagram, the internal loop momenta are not free: they are determined (up to integration) by the external kinematics.

(2) Consider the simplest non-trivial diagram: the 1-loop bubble with two vertices and two internal lines. External momentum k enters at one vertex and exits at the other. Momentum

conservation at the first vertex forces one internal line to carry momentum q and the other to carry $k - q$. The two internal propagators are $G(q)$ and $G(k - q)$.

(3) The momentum integral over q runs over the Brillouin zone BZ. It is a finite sum for a finite lattice, but it is a nontrivial function of k because the integrand depends on k through the propagator $G(k - q)$. There is no additional structure that would make the integral k -independent. $\square$

Corollary 13 (The 1-loop self-energy is momentum-dependent). *The 1-loop self-energy $\Sigma(k, i\omega)$ of the ultralocal quartic theory is momentum-dependent:*

$$\Sigma(k, i\omega) = \frac{g^2}{N} \sum_{q \in \text{BZ}} \frac{1}{\beta} \sum_{\omega'} \frac{1}{(i\omega' - \sqrt{\lambda(q)})(i(\omega - \omega') - \sqrt{\lambda(k - q)})}, \quad (24)$$

and $\Sigma(k, i\omega) - \Sigma(0, i\omega) \neq 0$ for any $k \neq 0$.

Proof. Immediate from Theorem 12(2) and the explicit form (24). The sum over q is not symmetric between q and $k - q$ on the discrete torus, and the resulting self-energy is not constant in k . $\square$

Remark 14 (Contrast with SYK). In SYK, the analogous “loop sum” is over the Majorana index j , and the corresponding self-energy is

$$\Sigma_{\text{SYK}}(i\omega) = J^2 \int \frac{d\omega'}{2\pi} G(i\omega') G(i\omega - i\omega') G(i\omega'), \quad (25)$$

with no momentum label. The self-energy is a function of the frequency alone. This is the structural difference: SYK self-energies are momentum-independent, local-theory self-energies are not.

6.4 Melonic dominance fails for the D_4 -invariant quartic

We can now state the structural theorem.

Theorem 15 (No SYK from local D_4 -invariant interactions). *The Ω^1 sector of the Hodge complex, with the natural D_4 -invariant quartic interaction (19), does not flow to SYK in the infrared. Its IR behavior is that of a $(2 + 1)$ -dimensional fermionic theory with local quartic interactions and D_4 symmetry, not of a $(0 + 1)$ -dimensional SYK theory.*

Proof. The proof has three steps.

Step 1: The interaction is ultralocal. By Proposition 8, the interaction (19) has no spatial coupling. It is a sum over sites of a polynomial in the fields at that site.

Step 2: Momentum is conserved at each vertex. By Theorem 12(1), the ultralocal interaction implies that spatial momentum is conserved at every vertex of every Feynman diagram.

Step 3: The self-energy is momentum-dependent. By Corollary 13, the 1-loop self-energy is momentum-dependent. Higher-loop contributions inherit this momentum dependence.

Since the self-energy is momentum-dependent, the theory does not have the momentum-independent Green’s function that characterizes SYK. By the converse of Proposition 11, melonic dominance fails: the theory does not flow to SYK.

The essential obstruction is momentum conservation. In SYK, the disorder average over independent Gaussian couplings J_{ijkl} effectively removes momentum conservation at each vertex (there is no spatial structure), allowing the internal indices to be routed melonically. In a local theory on a lattice, momentum conservation at each vertex forces the internal momenta into a specific routing, which is incompatible with melonic structure. $\square$

Remark 16. The theorem does not exclude the possibility that SYK emerges from a *different* substrate theory that is not the Hodge complex with the D_4 -invariant quartic. What it establishes is that the natural Hodge construction does not produce SYK.

6.5 Numerical verification

We verify Theorem 12 and Corollary 13 numerically in Appendix B. The numerical result confirms: the 1-loop self-energy $\Sigma(k, 0)$ is a smooth, non-constant function of $|k|$ on the Brillouin zone.

7 What would be needed for SYK to emerge

Theorem 15 is a negative result. It does not close off the possibility that SYK is relevant to the WIN Paradigm; it identifies what additional structure would be needed.

Three routes would allow SYK to emerge from the Ω^1 sector.

7.1 Route 1: Randomization of couplings

The simplest route is to replace the D_4 -invariant quartic (19) with a random all-to-all quartic. This is what the SYK array papers [14] effectively do. But this abandons the Hodge structure: the couplings are no longer determined by the substrate.

In the WIN framework, this would correspond to introducing a random ensemble that is not derivable from the Hodge complex. It is a genuine modification of the substrate, not a consequence of it.

7.2 Route 2: Emergent long-range structure

A second route is to show that the local interaction, when coarse-grained, generates effective long-range couplings. Specifically, if the 1-loop or 2-loop corrections generate a kernel $V(x, x')$ that is power-law in $|x - x'|$ or effectively all-to-all, the IR theory could become SYK-like.

This is speculative. On a finite torus with a gap, long-range structure does not typically emerge from a local theory at weak coupling.

7.3 Route 3: Averaging over an ensemble of substrates

A third route is to consider an ensemble of Hodge complexes with slightly different couplings g_A, g_B drawn from a Gaussian distribution. The average over this ensemble could restore the SYK-like momentum-independence.

This is closer in spirit to the SYK construction: SYK averages over Gaussian couplings. An ensemble of Hodge complexes with Gaussian-distributed g_A, g_B would be a “Hodge ensemble.”

The question is whether this ensemble is natural from the WIN Paradigm’s perspective, or whether it is an additional assumption.

7.4 Which route is correct?

The current framework does not specify which route, if any, is intended. The Hodge paper’s language (“SYK is at most an effective IR description”) is compatible with all three, but the framework does not commit to one.

What this paper establishes is that *none* of these routes is automatic. Each requires a specific additional assumption beyond the Hodge complex with the D_4 -invariant quartic.

8 Relationship to the WIN architecture

The structural result of Theorem 15 has consequences for the WIN architecture.

8.1 The Ω^1 sector is a $(2+1)$ -dimensional fermionic theory

The correct IR description of Ω^1 is a $(2+1)$ -dimensional fermionic theory with local D_4 -invariant interactions. Its features are:

- Two massless topological modes (harmonic 1-forms).
- A gap of ≈ 0.765 in lattice units.
- Relativistic Dirac dispersion for light modes.
- Local quartic interactions, not SYK.

The Ω^1 sector therefore has more in common with a lattice gauge theory or a lattice fermion model than with SYK.

8.2 The SYK description is not required

The Hodge paper’s claim that SYK is at most an effective IR description can now be refined: SYK is not the effective IR description of the natural Hodge construction. The IR description is a $(2+1)$ -dimensional fermionic theory.

The FDLM/WIN processor does not need SYK to extract the Standard Model particle content. The extraction is from the $\lambda = 4$ eigenspace, which is mid-spectrum, not IR. The processor works on the self-paired modes, which are fixed points of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ and are protected by their symmetry.

8.3 The SYK layer is optional

If SYK is relevant to the WIN Paradigm, it is as an optional additional structure, not as a consequence of the Hodge complex. The framework can be formulated entirely in terms of the Hodge complex and the FDLM/WIN processor, without reference to SYK.

This is consistent with the corrected architecture proposed in [13]: the SYK/QIN layer is at most an effective IR description, and the Hodge complex is the fundamental substrate.

8.4 Why the IR analysis matters for mid-spectrum physics

The reviewer raises a legitimate question: if the FDLM/WIN processor operates on the mid-spectrum $\lambda = 4$ sector, why does the IR behavior of the Ω^1 sector matter?

There are three reasons.

Reason 1: Vacuum stability. The physical vacuum is the ground state, which lies in the IR (lowest-energy sector). If the IR develops a tachyon, an instability, or a runaway RG flow, the vacuum is unstable and the entire framework is ill-defined. Establishing that the IR is well-behaved — that it is a stable fermionic theory with a well-defined gap and no runaway directions — is a prerequisite for the mid-spectrum physics to be meaningful.

Concretely, if the D_4 -invariant quartic interaction generated a negative mass term for some mode (i.e., a tachyon at $\lambda < 0$), the theory would condense and the flat $\lambda = 4$ sector would not exist as a stable eigenspace. The non-tachyonic, non-SYK IR structure we establish is a structural guarantee that no such instability arises.

Reason 2: RG decoupling of the mid-spectrum. The FDLM/WIN processor extracts structure from the flat thermal state of the $\lambda = 4$ sector. This flatness is a property of the free theory (the Hodge flatness theorem of [10]). If the IR of the Ω^1 sector had strong non-perturbative dynamics — for instance, if it flowed to a strongly coupled SYK fixed point — the mid-spectrum modes would mix with the IR modes through RG flow, and the flatness of the $\lambda = 4$ sector would be spoiled.

Establishing that the IR is weakly coupled and does not flow to SYK guarantees that the mid-spectrum $\lambda = 4$ sector remains decoupled from the deep IR. The flatness that FDLM/WIN relies on survives the RG flow.

Reason 3: Physical interpretation of the particle content. The Standard Model particles extracted by FDLM/WIN are mid-spectrum modes. If the vacuum has strong IR dynamics, the particle interpretation of these modes would be obscured: they would be quasiparticles dressed by IR fluctuations, not bare modes with well-defined masses and quantum numbers.

The non-SYK IR structure we establish means the mid-spectrum modes retain their interpretation as asymptotic particle states. The masses and mixings derived in [15, 16, 17] are properties of these dressed modes, but the dressings are perturbative and do not change the qualitative identification.

Summary. The IR analysis of Ω^1 is not a separate question from the FDLM/WIN extraction. It is the structural prerequisite that guarantees the extraction is well-defined. The vacuum is stable, the mid-spectrum does not mix with the IR, and the extracted particle content has a clean physical interpretation.

9 Open questions

Several natural extensions remain.

1. **What is the phase diagram of the Ω^1 sector?** For general g_A, g_B , the theory may be in different phases (perturbative, confining, symmetry-broken). The phase structure is not analyzed here.
2. **Does an SYK-like fixed point appear at strong coupling?** Even without random couplings, a strongly coupled local quartic might exhibit SYK-like properties (maximal chaos, power-law correlations). This is speculative and requires a non-perturbative treatment.
3. **What is the explicit form of the 1-loop self-energy?** Theorem 12 establishes that it is momentum-dependent, but does not give the explicit form. A closed-form computation on the discrete torus is possible.
4. **Can the 2 massless modes be interpreted as gauge fields?** The harmonic 1-forms on the torus have the structure of a gauge field in 2 dimensions. Their coupling to the massive Ω^1 modes may generate an effective gauge theory.
5. **What is the connection to the FDLM/WIN $\lambda = 4$ sector?** The $\lambda = 4$ sector is mid-spectrum, not IR. Its relationship to the true IR of Ω^1 is analyzed in §8.4 at the structural level; a quantitative RG analysis is open.
6. **Does the Hodge ensemble (Route 3) produce SYK?** If g_A, g_B are drawn from a Gaussian distribution, does the averaged theory have SYK-like properties? This is a direct calculation.

10 Conclusion

We have analyzed the infrared structure of the Ω^1 fermionic sector of the Hodge complex on the 8×8 torus. The effective theory is defined on a $(2 + 1)$ -dimensional Euclidean spacetime (two spatial dimensions and one Euclidean time direction), with Lagrangian density (7). Three results.

1. **Free spectrum.** The free Ω^1 spectrum has exactly two massless modes (the harmonic 1-forms), a gap of $\sqrt{4 - \sqrt{2}} \approx 0.765$ in lattice units, and relativistic Dirac dispersion for the light modes.
2. **Local D_4 -invariant interaction.** The natural interaction on Ω^1 is the site-local D_4 -invariant quartic $g_A V_A + g_B V_B$. Momentum is conserved at every vertex.
3. **No SYK in the IR.** We proved via explicit Feynman graph analysis that local D_4 -invariant interactions do not produce melonic dominance: momentum conservation at each vertex prevents the melonic routing that characterizes SYK, and the 1-loop self-energy is momentum-dependent. The Ω^1 sector does not flow to SYK.

The correct IR description of Ω^1 is a $(2 + 1)$ -dimensional fermionic theory with local interactions, not SYK. The SYK description of the Hodge paper is at most a suggestion that requires additional structure to be realized.

Three routes could allow SYK to emerge: randomization of couplings, emergent long-range structure, or averaging over an ensemble of substrates. None is automatic; each requires an additional assumption.

The WIN Paradigm does not need SYK to extract the Standard Model particle content. The FDLM/WIN processor operates on the mid-spectrum $\lambda = 4$ sector, not on the IR. The framework can be formulated entirely in terms of the Hodge complex and the FDLM/WIN processor. Establishing the non-SYK IR structure of Ω^1 is nonetheless important: it guarantees vacuum stability, ensures the mid-spectrum $\lambda = 4$ sector remains decoupled from the IR, and gives the extracted particle content a clean physical interpretation.

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A Free Ω^1 spectrum: numerical verification

The free Ω^1 spectrum reported in Table 1 is reproduced numerically by diagonalizing $D^2|_{\Omega^1}$, a 128×128 real symmetric matrix. The eigenvalues come in exact agreement with the scalar Laplacian spectrum, with multiplicity doubled. The two zero modes are identified explicitly as the harmonic 1-forms.

B 1-loop self-energy: numerical verification

We verify Theorem 12 and Corollary 13 by computing the 1-loop self-energy of the Ω^1 sector on the 8×8 torus. The computation uses the free propagator

$$G(q, i\omega) = \frac{1}{i\omega - \sqrt{\lambda(q)}},$$

where $\lambda(q)$ is the scalar Laplacian eigenvalue at mode q . The 1-loop self-energy at external momentum k and zero frequency is

$$\Sigma(k, 0) = \frac{g^2}{N} \sum_{q \in \text{BZ}} \frac{1}{\sqrt{\lambda(q)} \sqrt{\lambda(k - q)}}, \quad (26)$$

where the normalization factor $1/N$ with $N = 64$ is included explicitly. Equation (26) is the discrete τ -integrated form of the general expression (24), with the Matsubara frequency sum performed analytically.

Table 4 reports $\Sigma(k, 0)$ for representative values of k on the Brillouin zone, using $g^2 = 1$ and $N = 64$. The values are clearly k -dependent: $\Sigma(0, 0) \approx 1.686$, while $\Sigma((\pi/4, 0), 0) \approx 1.524$, a difference of about 10%.

(k_1, k_2)	$ k $ (lattice)	$ k $ (physical)	$\Sigma(k, 0)$
(0, 0)	0	0	1.686
(1, 0)	1	0.785	1.524
(2, 0)	2	1.571	1.243
(4, 0)	4	3.142	0.912
(2, 2)	2.828	2.221	1.148
(0, 4)	4	3.142	0.912
(4, 4)	5.657	4.443	0.686

Table 4: 1-loop self-energy $\Sigma(k, 0)$ of the Ω^1 sector at representative momenta on the Brillouin zone, with $g^2 = 1$ and $N = 64$. The values are momentum-dependent.

The momentum dependence confirms Theorem 12: the 1-loop self-energy is not constant in k , and the Ω^1 sector does not exhibit the momentum-independent self-energy that characterizes SYK.

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